

ECON 1020: Principles of Microeconomics

Homework 2: Chapter 3

Interdependence and the Gains from Trade

Due on Monday, September 21 at 12:20 PM

Submissions are accepted either online through NTU COOL or as a hard copy at your recitation session.

Part A: Interdependence and the Gains from Trade

1. Suppose that Taiwan produces semiconductors and rice, and it can switch production between the two at a constant rate. If the most semiconductors it can produce is 300 (in some unit) and the most rice it can produce is 50,000 tons, what is the opportunity cost of one unit of semiconductors, and what is the opportunity cost of one ton of rice?

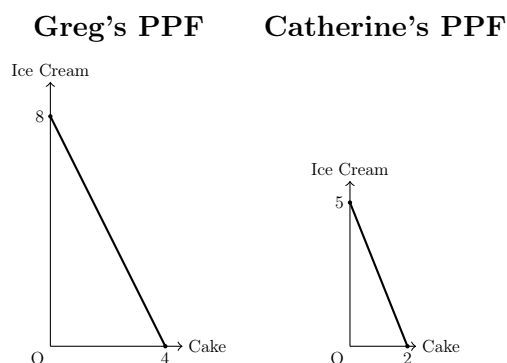
2. In Country A, a worker who works 40 hours can produce 200 pounds of rice or 100 pounds of broccoli. In Country B, a worker who works 40 hours can produce 160 pounds of rice or 120 pounds of broccoli.
 - (a) Which country, if either, has an absolute advantage producing rice? Producing broccoli?

 - (b) Which country has a comparative advantage producing rice? Producing broccoli? Defend your answer using the numbers given.

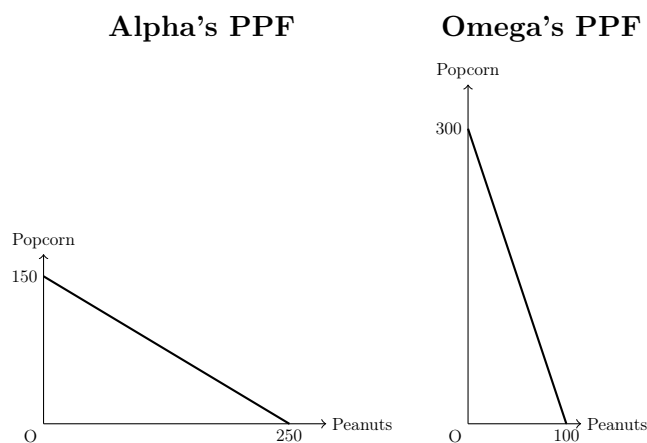
 - (c) Give a range of prices, in pounds of rice per pound of broccoli, at which the two countries would both be willing to trade.

3. Suppose that Taiwan can produce 5 smart displays per hour of labor and Country X can produce 3 smart displays per hour of labor. Does it necessarily follow that Taiwan should specialize in smart display production? Explain your answer using the concepts of absolute and comparative advantage.

4. The production possibilities frontiers below show how much Greg and Catherine can each produce in 8 hours of time.



- (a) What is Greg's opportunity cost of producing a cake? What is Catherine's?
- (b) Which of them has an absolute advantage in producing ice cream? In producing cake?
- (c) Which of them has a comparative advantage in producing ice cream? In producing cake?
- (d) Is it possible for Greg and Catherine to gain from trade? Defend your answer.
5. Alpha and Omega are the only two countries in the world, and each faces the production possibilities frontier shown below.



- (a) Suppose each country uses half of its resources to produce each good. On each graph above, plot and label this point **A**.
- (b) If the two countries do not trade, what is the total world production of popcorn and of peanuts?
- (c) Now suppose each country specializes completely in the good in which it has a comparative advantage. What is the total world production of each good now?
- (d) Suppose each country then trades 100 units of popcorn for 100 units of peanuts. On each graph above, plot and label the resulting consumption point **B**, and explain why point B shows a gain from trade.
6. Zimbabwe and Portugal can each switch between producing toothbrushes and hairbrushes at a constant rate. The table below shows the machine-minutes needed to produce one unit of each good. Each country has 60 minutes of machine time available per day.

Machine minutes per unit	Toothbrush	Hairbrush
Zimbabwe	3	10
Portugal	5	6

- (a) Draw each country's production possibilities frontier.

(b) What is each country's opportunity cost of a toothbrush? Of a hairbrush?

(c) Which country has the comparative advantage in toothbrushes? In hairbrushes?